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Paleomagnetism of the Chichibu Quartz Diorite—Constraints on the Time of Lateral Bending of the Kanto Syntaxis

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Paleomagnetic directions have been determined on Late Miocene diorite bodies in the Kanto Mountains in an effort to clarify the timing of lateral bending of the Kanto Syntaxis. A mean paleomagnetic direction (D=3.5°, I=64.4°, α_{95}=5.1°) of the Chichibu Quartz Diorite, obtained from 21 samples, shows no significantly deflected declination. This indicates no tectonic rotation of the Kanto Mountains since intrusion of the Chichibu Quartz Diorite (6–8 Ma). This result signifies that the cusp in the pre-Neogene zonal structure (Kanto Syntaxis) was formed in the Late Miocene or earlier.

1. Introduction

The pre-Neogene geologic terranes of the Japanese Islands are characterised by a zonal structure (Fig. 1). This zonal structure is sharply bent at the Fossa Magna,

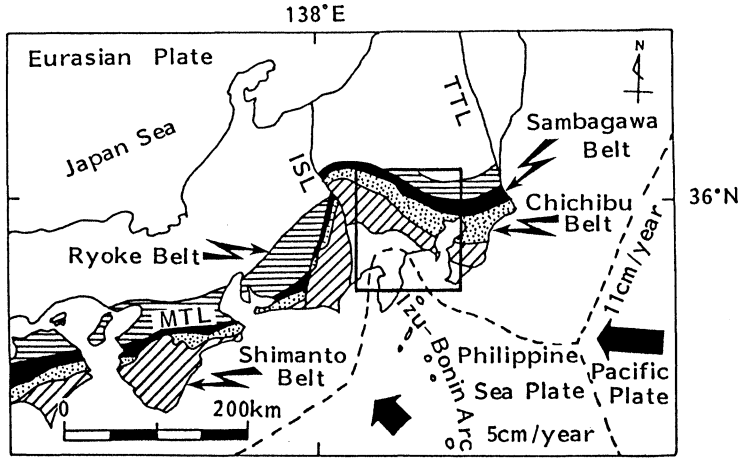


Fig. 1. Zonal structure of pre-Neogene Terranes and cusp-shape bending form at South Fossa Magna. MTL: Median Tectonic Line, ISL: Itoigawa-Shizuoka Tectonic Line, TTL: Tanakura Tectonic Line, arrow: present plate motion relative to the Eurasian Plate (MINISTER and JORDAN, 1978, 1979).

central Honshu, and extends west to the Kyushu district. This feature was noticed in the last century (NAUMANN, 1885; HARADA, 1888). The timing of this cusp formation is significant in the tectonic development of the Japanese Islands.

KOBAYASHI (1941) explained this bend in the Cretaceous deformation of Honshu by the existence of an obstacle in front of the southward migrating Honshu. TATEIWA (1976) considered that this cusp had formed in the Cretaceous to Paleogene time. MATSUDA (1978, 1984) also placed the time of this cusp formation in the pre-Miocene, while NIITSUMA (1982) set forth that the subduction of the Philippine Sea Plate initiated at 6–7 Ma in the Late Miocene, and resulted in the bending of the zonal structure. Moreover, NIITSUMA and MATSUDA (1985) gave one explanation that the cusp was formed by collisions of the Tanzawa and Izu Blocks, on the basis of the exotic origin of these blocks and past plate motion. Further, HYODO and NIITSUMA (1986) referred mean paleomagnetic directions of the Chichibu Tertiary System to clockwise rotation of the Kanto Mountains at about 94° since the Early Miocene. They explained this rotation by the opening of the Japan Sea (45°, 15 Ma; OTOFUJI *et al.*, 1985) and by the collision of the Tanzawa Block (45°, after 6–7 Ma; NIITSUMA, 1982). On the other hand, ITOH (1988) carried out a paleomagnetic study on Late Cretaceous and Miocene rocks in the eastern part of Southwest Japan, and concluded that the bending motion occurred between 15 and 12 Ma. Thus several timings have been considered for the cusp formation.

The Kanto Mountains are on the east wing of the cusp, and the Chichibu Quartz Diorite intrudes into the central part of the Kanto Mountains (Fig. 2). The radiometric ages of this quartz diorite determined by UENO and SHIBATA (1986) all indicate the Late Miocene (6–8 Ma). We measured paleomagnetism of the Chichibu Quartz Diorite and concluded that the cusp in the zonal structure had been formed in the Late Miocene or earlier.

2. Geologic Setting

The geologic map of the Kanto Mountains and adjacent area is shown in Fig. 2. Pre-Neogene zonal structures trending WNW-ESE comprise the Sambagawa Metamorphics, the Chichibu Belt, the Sanchu Graben and the Shimanto Belt. The Chichibu Quartz Diorite is in Nakatsugawa, about 20 km west of Chichibu City. Sedimentary rocks in this area are the Nakatsugawa Group (FUJIMOTO *et al.*, 1950), which is comprised of the Ishibune, Ryogami and Okawamata Formations, in ascending order (FUJIMOTO *et al.*, 1957; ISHII, 1962; OKUBO and HORIGUCHI, 1969), and consists of massive sandstone, shale, limestone and chert. These strata generally strike ESE and dip steeply NNE and are disrupted by WNW-ESE trending thrust faults.

Igneous rocks intruding into the Nakatsugawa Group are stocks of medium-grained massive quartz diorite, and small dykes of quartz porphyry and porphyrite.

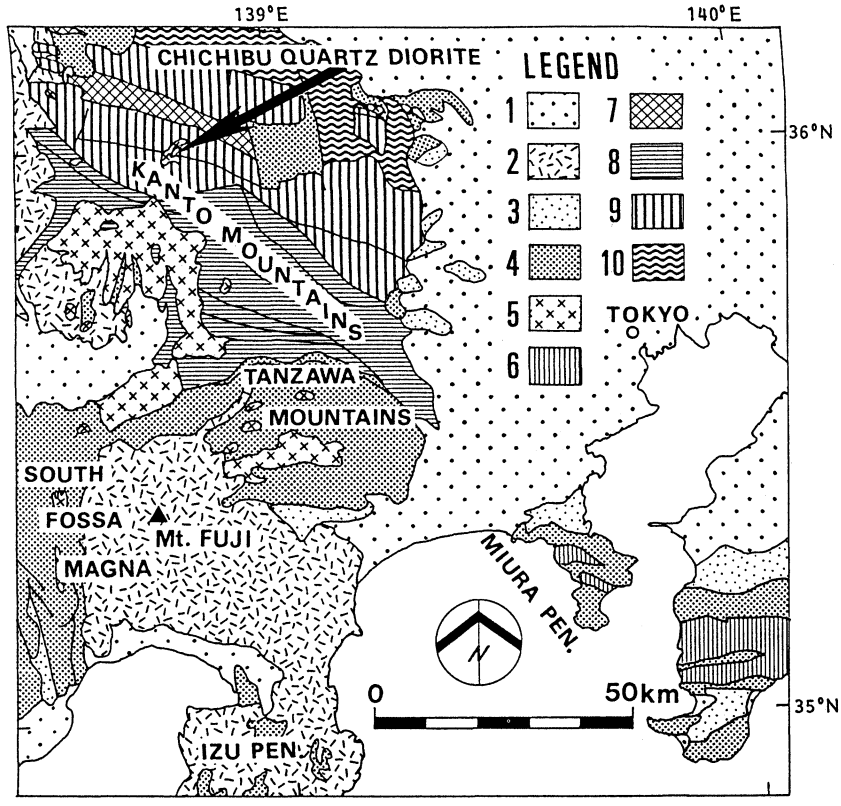


Fig. 2. Geologic map of the Kanto Mountains and its adjacent area, simplified from YAMADA *et al.* (1982). 1: Pleistocene deposits, 2: Pliocene-Pleistocene volcanic rocks, 3: Pliocene deposits, 4: Middle-Late Miocene deposits, 5: Granitic rocks, 6: Paleogene deposits, 7: Sanchu Graben, 8: Shimanto Belt, 9: Chichibu Belt, 10: Sambagawa Belt.

3. Sampling and Measurement

Oriented hand samples were collected from the Chichibu Quartz Diorite (Fig. 3). More than three cylindrical specimens were cored from each of the oriented hand samples in the laboratory. The total number of samples and specimens are 51 and 304, respectively. Remanent magnetization was measured with a Schonstedt spinner magnetometer at Gunma University. After measuring the natural remanent magnetization (NRM) of all specimens, a pilot specimen from each sample was progressively demagnetized in an alternating field (AF) up to 40 mT at increments of 10 mT. The AF demagnetizer has a three-axis tumbler, and a Helmholtz coil system for cancellation of the ambient field.

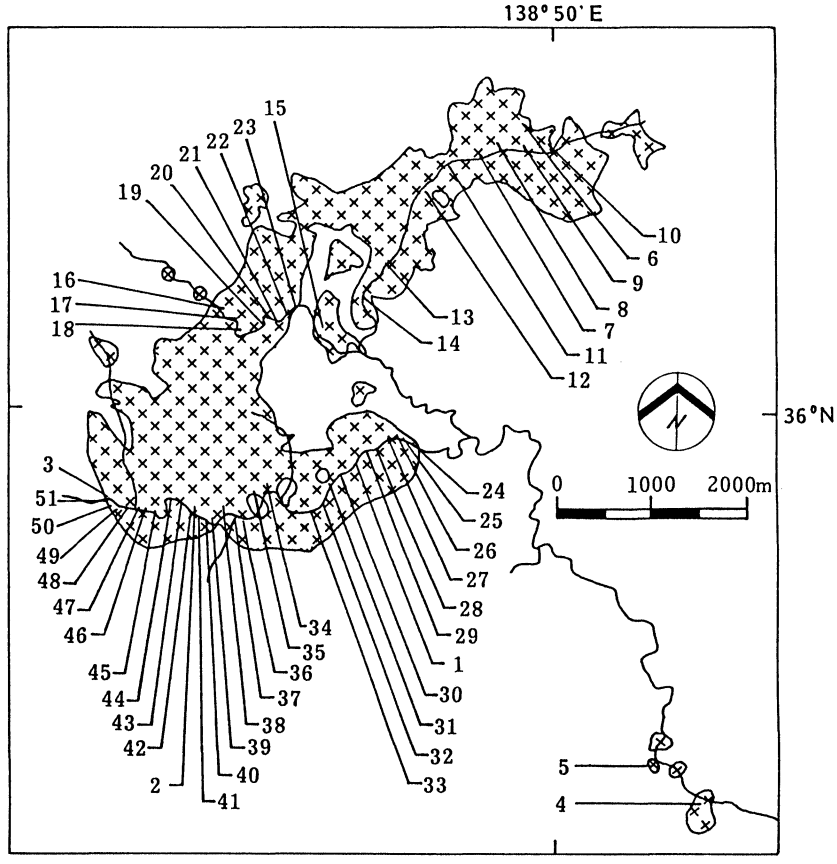


Fig. 3. The localities of paleomagnetic samples.

4. Results

By the progressive AF demagnetization, pilot specimens are sorted into three categories (Fig. 4): (A) remanent magnetizations are stable with characteristic directions (Fig. 4(a)); (B) remanent magnetizations turn to characteristic directions (Fig. 4(b)); and (C) remanent magnetizations change direction irregularly and relative intensities decrease rapidly through AF demagnetization (Fig. 4(c)). Secondary magnetic overprints can be effectively erased by AF demagnetization of the (A) and (B) categories. In particular, the relative intensities of specimens with reversed polarity increased after AF demagnetization. However, as for the specimens in the third category, we could not obtain stable magnetizations, and thus rejected 30 samples.

At each sample, scattered NRM directions of the specimens in (A) and (B)

categories make tight clusters after demagnetization at 20 mT (Figs. 4(a) and 4(b)). Alpha 95 values (FISHER, 1953) are less than 10° for every hand sample, which indicates homogeneous magnetization within each sample. The mean magnetic directions of each sample and whole rock body after demagnetization at 20 mT are shown in Table 1.

The paleomagnetic directions of the Chichibu Quartz Diorite are shown in Fig. 5, which clearly shows that paleomagnetic declinations (D=3.5°) have no significant deflection from the north. On the other hand, paleomagnetic inclinations (I=64.4°) are about 9° larger than that of the geocentric axial dipole value (55.5° N).

5. Interpretation and Discussion

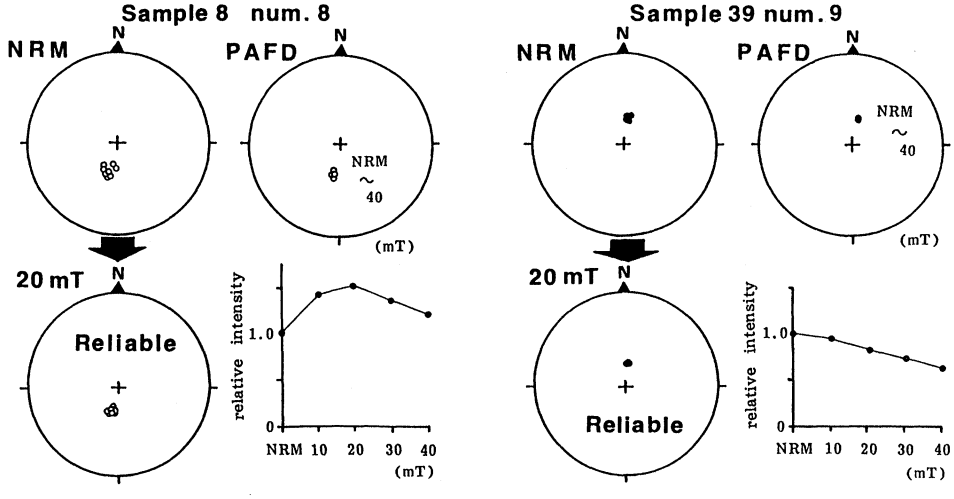
Paleomagnetism of the northeastern part of the Chichibu Quartz Diorite shows reversed polarity. On the other hand, its southwestern part includes both normal and reversed polarity: reversed polarity is recognized only near the margin of the diorite body. The lithofacies of the Chichibu Quartz Diorite is not significantly heterogeneous, implying a monogenic origin. Hence, the opposing polarity is possibly due to a change in geomagnetism during the period of cooling of the quartz diorite.

Mean paleomagnetic directions of the 21 reliable samples are D=3.5°, I=64.4° and α_{95}=5.1° after inversion of the reversed polarity directions to normal polarity. Paleomagnetic directions are distributed in antipodal directions. These 21 samples include both normal polarity (13 samples) and reversed polarity (8 samples), so the measured directions represent magnetic fields of a sufficiently long period to average geomagnetic secular variations. The paleomagnetic directions of the Chichibu Quartz Diorite can be assumed to represent an average dipole field.

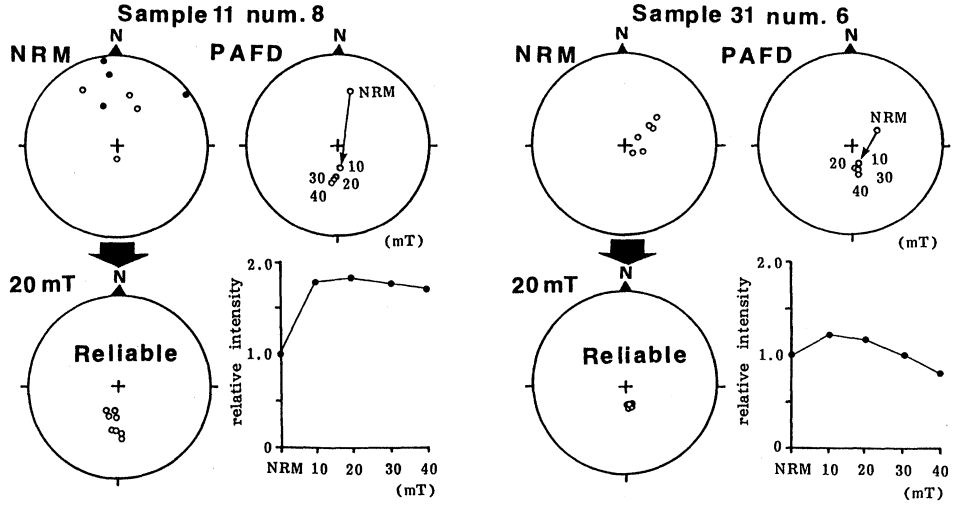
The paleomagnetic direction shows no significantly deflected declination from the north, so the Kanto Mountains have not rotated since intrusion of the quartz diorite. The radiometric ages of the Chichibu Quartz Diorite (6–8 Ma) indicate the Late Miocene (UENO and SHIBATA, 1986). Therefore, we conclude that the Kanto Mountains, including the Chichibu Quartz Diorite, have not rotated since the Late Miocene.

Several timings are considered for the sharp bend formation of pre-Neogene zonal structure; the pre-Miocene, Middle Miocene and post-Late Miocene. Taking into account the evidence that Southwest Japan rotated clockwise at a 47° arc at 14.9 Ma, and that the duration of its rotation was as short as less than 1 m.y. (OTOFUJI *et al.*, 1985), as well as paleomagnetic data from the Chichibu Quartz Diorite, we conclude that the cusp in the zonal structure was formed in the Late Miocene or earlier. Furthermore, this result implies that it is hard to explain the cusp formation as a result of collisions of the Tanzawa and Izu Blocks, because the Kanto Mountains have not rotated since 6–8 Ma.

The mean inclination of the Chichibu Quartz Diorite (I=64.4°) is about 9° larger than the geocentric axial dipole value (55.5° N). This deviation may not be related to north-south translation, because the Kanto Mountains have not changed

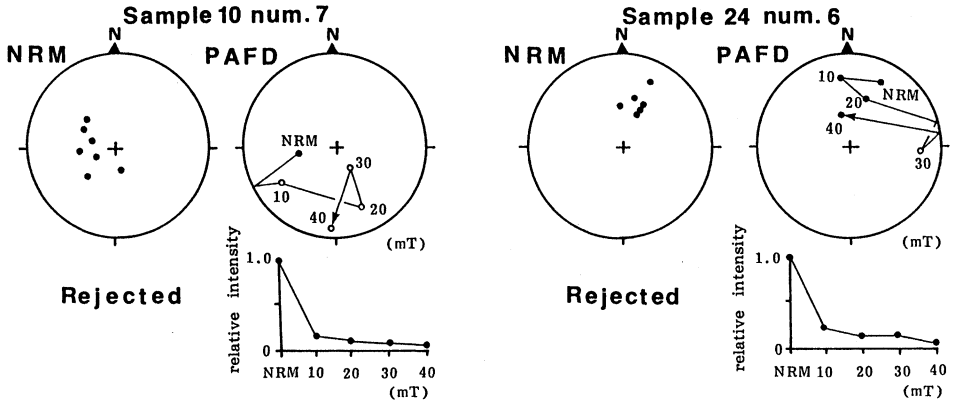


(a)



(b)

Fig. 4. Typical demagnetization curves through progressive alternating field demagnetization (PAFD). Magnetically stable ((a) and (b): Samples 8, 39, 11 and 31) and unstable ((c): Samples 10 and 24) samples. Projections are on equal area, solid symbols on the lower hemisphere, open symbols on the upper hemisphere. (a), (b) Magnetization is regarded as characteristic component, except the soft secondary component removed at 10 mT. (c) Secondary component cannot be removed by alternating field demagnetization.



(c)

Fig. 4. (continued).

Table 1. Reliable remanent magnetizations of the Chichibu Quartz Diorite.

Sample	N	M (A/m)	AFD (mT)	D(°)	I(°)	α_{95}	k	VGP long.	VGP lat.
1	7	0.34	20	− 21.0	72.5	8.1	56.0	112.8°E	64.2°N
2	8	0.29	20	34.4	76.5	1.8	964.6	164.0°E	55.0°N
5	6	0.035	20	170.4	− 57.4	5.8	132.4	66.4°E	82.1°N
8	8	0.77	20	− 170.0	− 69.9	1.9	900.6	157.1°E	70.9°N
11	8	0.11	20	− 173.0	− 55.6	9.1	37.8	134.7°W	84.3°N
12	9	0.065	20	− 154.8	− 38.9	3.0	294.7	105.4°W	64.0°N
14	7	0.070	20	− 170.8	− 59.6	5.3	129.9	165.5°W	81.5°N
15	4	0.20	20	− 41.4	50.5	1.0	9306.9	53.1°E	55.5°N
26	6	0.20	20	0.5	76.9	8.0	71.9	139.2°E	61.0°N
27	5	0.31	20	28.4	70.5	3.1	593.0	176.1°E	63.0°N
28	8	0.69	20	3.3	65.8	1.8	921.5	149.2°E	77.7°N
30	6	0.34	20	1.8	66.6	4.1	270.2	144.0°E	76.8°N
31	6	0.17	20	171.3	− 71.8	1.1	3459.5	125.7°E	68.5°N
32	5	0.14	20	8.4	70.7	1.2	4128.3	153.1°E	70.1°N
36	6	0.015	20	11.7	75.6	5.2	164.6	150.3°E	62.2°N
38	3	0.32	20	− 1.0	57.7	3.5	1262.2	120.3°E	87.5°N
39	9	0.14	20	12.2	67.4	1.4	1346.5	166.8°E	73.3°N
42	5	0.11	20	18.9	61.8	1.6	2392.6	162.7°W	73.9°N
43	4	0.067	20	9.3	51.6	5.1	326.9	107.9°W	81.4°N
50	5	0.29	20	176.2	− 64.7	6.4	144.1	125.0°E	79.0°N
51	9	0.47	20	174.3	− 48.1	6.7	59.4	4.7°W	81.6°N
Total	134		20	3.5	64.4	5.1	40.2	152.1°E	79.4°N

N: number of specimens, M: mean intensity before demagnetization in A/m, AFD: demagnetization level in mT, D, I: sample-mean declination and inclination of remanent magnetization, α_{95}: radius of the 95% confidence circle (FISHER, 1953), k: precision parameter, long, lat: longitude and latitude of virtual geomagnetic pole position (northern hemisphere).

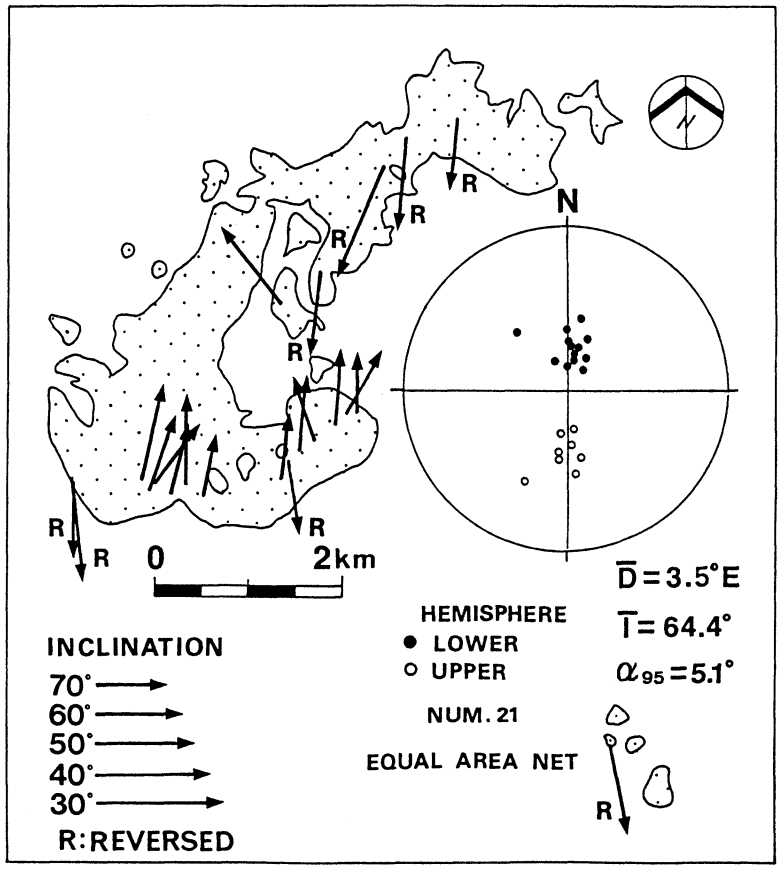


Fig. 5. Characteristic paleomagnetic directions of the Chichibu Quartz Diorite.

their latitude since the Early Miocene (HYODO and NIITSUMA, 1986); the most probable cause is tilting of the Chichibu Quartz Diorite.

A geologic block diagram around the Kanto Mountains and the simplest model of tilting are shown in Fig. 6. Pre-Neogene formations are cut by many thrust faults, which are parallel to the zonal structures (110° clockwise from the north) and dip steeply (60~70°) NNE. Consequently, we can develop a basic model of the tilting motion, as shown in Fig. 6. Fault blocks ("dominoes") tilt to the south, and each fault dips NNE. After the fault blocks rotated under N-S compression, the angle of the thrust faults increased, and each domino rose. When we assume that the axis of tilting motion was parallel to the strike of the thrust fault, a tilting motion (θ) of 10° brings about the mean inclination of the Chichibu Quartz Diorite we observed, because of the geocentric axial dipole value of 55.5°. Assuming no ductile deformation takes place within each domino, we can find a shortening factor (d), as

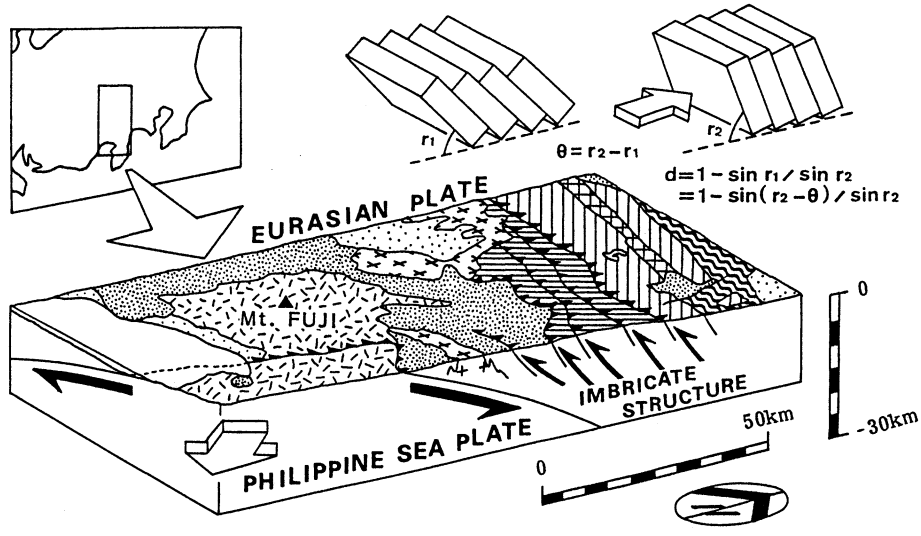


Fig. 6. Geologic block diagram of the Kanto Mountains and its adjacent area and the simplest tilting model of "dominoes". Pre-Neogene formations (Chichibu and Shimanto Belts) of the Kanto Mountains are regarded as an accretionary prism (HISADA, 1983; HISADA and KISHIDA, 1986) and they are cut by many thrust faults, which are parallel to zonal structures (WNW-ESE) and dip steeply NNE. These fault blocks ("dominoes") rose as a result of rotation during N-S compressional deformation associated with the subduction of the Philippine Sea Plate.

we know the angles of the present fault-plane dip (r_2) and the tilting motion (θ) of these dominoes:

$$d = 1 - \sin(r_2 - \theta)/\sin r_2.$$

Here, we can obtain the value of θ using paleomagnetic directions, because the mean inclination of the Chichibu Quartz Diorite suggests tilting of about 10° toward NNE. Consequently, we find a shortening value (d) of 7.8~11.5% for the Kanto Mountains since the Late Miocene, when we assume the angle of the thrust fault dip (r_2)=60~70°. Though the above is an ideal case and only one of the probable explanations, an increase in inclination can occur in this way.

On the basis of geological evidence, MATSUDA (1978) proposed that the Izu Peninsula had collided with central Honshu in the Early Quaternary and then moved northward for a few tens of kilometers. YOSHIDA *et al.* (1984) inferred a clockwise rotation of the Miura Peninsula, about 50 km northeast of the Izu Peninsula, as large as 30° since 4 Ma. They considered that its rotation was associated with collisions of the Izu Peninsula, and estimated about 30 km for the northward component of post-collision movements of the Izu Peninsula. Moreover, NIITSUMA and MATSUDA (1985) considered that the Tanzawa and Izu Blocks had

collided with central Honshu since the Late Miocene. These tectonic processes of N-S compressional deformation are consistent with the tilting motions of the Kanto Mountains.

The mean declination of the Chichibu Quartz Diorite (D=3.5°) increases in accordance with the tilt correction mentioned above. Tilt-corrected declination of the Chichibu Quartz Diorite (D'=10°) may imply a clockwise rotation of the Kanto Mountains since the Late Miocene (6–8 Ma), though not enough to form the cusp in the pre-Neogene zonal structure.

6. Conclusion

1) Paleomagnetic directions of the Chichibu Quartz Diorite were determined for 21 samples. The mean direction (D=3.5°, I=64.4°, α_{95}=5.1°) shows no significantly deflected declination from the north.

2) The Kanto Mountains have not rotated since intrusion of the Chichibu Quartz Diorite (6–8 Ma), which leads to the conclusion that the cusp in the pre-Neogene zonal structure in the south Fossa Magna region was formed in the Late Miocene or earlier.

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